NIA Challenge Phase 3

SpeechCARE Team

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This report consists of 8 components:

Component 1 introduces a six-step pipeline for denoising, transcribing, diarizing, and segmenting clinical audio to enable accurate, scalable ADRD analysis. See notebook: preprocessing_scripts.ipynb

Component 2 explains SpeechCARE's modular system with four model variants that fuse acoustic, linguistic, and clinical data to detect cognitive decline, adaptable to diverse datasets, tasks, and resource constraints. See notebook: SpeehCARE_script.ipynb

Component 3 incorporates bias mitigation, noise robustness, and speaker verification to ensure fair, reliable, and secure use of SpeechCARE in real-world and longitudinal clinical settings. See notebook: Trustworthiness_script.ipynb

Component 4 assesses SpeechCARE's cross-lingual and cross-task performance on five DementiaBank corpora using zero-shot, transfer, in-corpus, and task-attentive strategies. See notebook: Generalizability_script.ipynb

Component 5 refines an interactive, multi-modal dashboard to explain model predictions via clinical, linguistic, and acoustic biomarkers—guided by clinician feedback to improve usability and interpretability. See: explainability.md

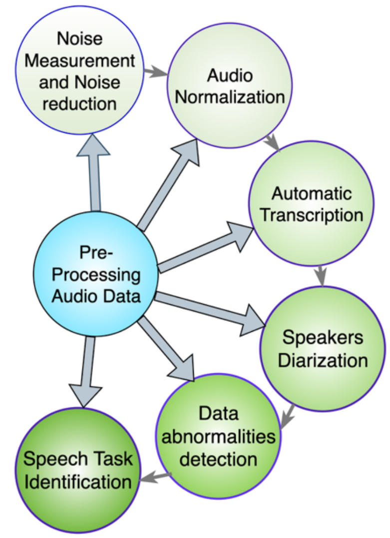
Component 6 introduces a multi-task learning pipeline that leverages related speech conditions to improve ADRD detection specificity, robustness, and generalizability through shared representation learning. See notebook: multi_task_learning_script.ipynb

Component 7 implements and evaluates continual learning with EWC and IMM to adapt SpeechCARE to new patient data and clinician feedback while preserving prior knowledge, enhancing real-world MCI detection. See notebook: continual_learning_script.ipynb

Component 8 Demonstrates SpeechCARE's effectiveness for passive, scalable MCI screening across home-based speech contexts—structured assessments, phone calls, and nurse interactions—achieving strong predictive performance in real-world clinical workflows. See notebook: Homecare_script.ipynb

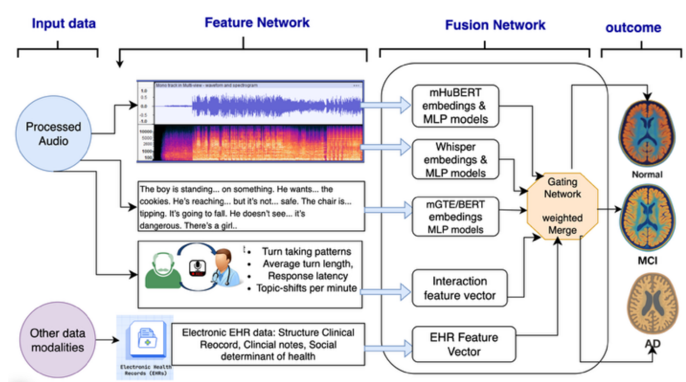
The integration of these components represents a novel approach in ADRD research that significantly enhances implementation feasibility in clinical settings for early MCI detection. This approach draws clinicians' attention to patients' cognitive functioning, facilitating timely evaluation and development of targeted interventions to mitigate adverse outcomes.

Figure 1. Preprocessing components



1. Pre-Processing Component and Performance (Figure 1). Real-world audio recordings often include background noise (e.g., traffic, air conditioners), speech from others (e.g., clinicians), and multiple speech tasks (e.g., story recall, picture description). To address these challenges, SpeechCARE's preprocessing component includes six steps: **(a) Noise Measurement & Reduction:** A low-pass[1] filter removes noise above 8 kHz, preserving critical prosodic features for cognitive decline; **(b) Audio Peak Normalization[2]:** Audio samples are scaled to their maximum amplitude, reducing variability caused by recording devices or microphone distance; **(c) Transcription:** Denoised and normalized recordings are transcribed using Whisper-large-v3[3] (publicly accessible, secure system for local servers) with Word Error Rate [4] (WER, accuracy of transcription) dropped from 0.1409 to 0.1370 (English), 0.4399 to 0.3564 (Spanish), and 2.7991 to 0.8608 (Mandarin) compared to Phase-2 results; **(d) Speaker Diarization**: A two-step pipeline uses WhisperX[5] for word-level timestamps, followed by Llama-3.1-405B[6] to diarize and tag each sentence as patient or clinician. On the benchmark Pitt corpus[7], it outperformed state-of-the-art model, NVIDIA NeMo[8] (WDER 0.09 vs 0.15) in diarization accuracy; **(e) Data Abnormality Detection:** To identify low-quality speech samples (e.g., missing patient-speech, dropped audio), LLaMA-3.3-70B[9] with prompt engineering and few-shot learning is used to detect the corrupted speech, achieving 100% precision on the Phase-2 dataset. **(f) Speech Task Identification:** LLaMA-3.3-70B, with prompt engineering, identifies speech tasks within long interviews and marks the start and end points of each task. This achieved 100% accuracy on the Delaware corpus[10] (Table 1). **IMPLICATIONS.** SpeechCARE enhances ADRD detection by improving transcription accuracy, filtering noise while preserving cognitive-relevant acoustic features, detecting poor-quality recordings, and automatically identifying speech tasks—all without manual review. Built on open-source, HIPAA-compliant[11] tools, it supports scalable, secure deployment on local servers.

Figure 2. SpeechCARE models



2. SpeechCARE Models (Figure 2) are a flexible, modular system for detecting early cognitive decline by analyzing speech, language, and patient clinical data. It includes two components: a Feature Network for extracting information from each input modality, and a Fusion Network that dynamically weights the contribution of each modality to predict Normal**,** MCI, or AD. SpeechCARE was implemented in four versions based on available data, study design, and computational resources**. (a) SpeechCARE-AGF (Adaptive Gating Fusion)** is the primary configuration. Raw audio is encoded using mHuBERT[12], a multilingual transformer pretrained on >90K hours of speech, to capture acoustic-temporal cues. Transcripts are processed with mGTE[13]**,** a multilingual linguistic transformer, to encode linguistic cues. Demographics, clinical, behavioral, and social risk factors are encoded as structured vectors. The Fusion Network assigns case-specific modality weights: for instance, it emphasizes acoustic features when voice instability is more informative than linguistic cues. This adaptability improves robustness to noise, task variability, and language diversity. **(b) SpeechCARE-Whisper** follows the same architecture but replaces mHuBERT with Whisper[14], a multilingual transformer trained on >600K hours of speech, processing mel-spectrograms, enabling joint extraction of acoustic and linguistic features from a single input. This version is suited for streamlined pipelines but may slightly reduce performance. **(c) SpeechCARE-Task-Attentive** is tailored for datasets with multiple speech tasks per participant (e.g., Delaware, Table 1). Each task—transcript, acoustic, or both—is encoded using the optimal pretrained transformer. An attention[15] network learns task relevance by weighting each task's contribution to the model's decision. This design enhances generalizability and improves MCI detection in multi-task settings. **(d) SpeechCARE-handcrafted** is designed for interpretability and low-resource environments. It relies on hand-crafted acoustic-temporal features (e.g., jitter, pauses, pitch), capturing diverse vocal attributes. It eliminates the need for large neural models, offering greater transparency for clinical interpretation.

Figure 3. Bias Mitigation Results Using Voice Conversion

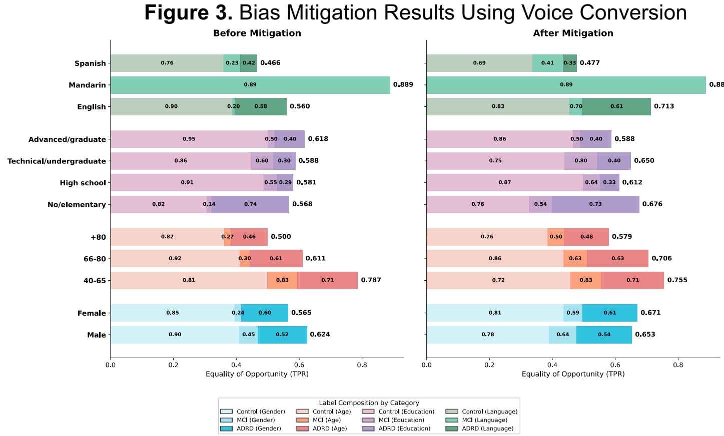
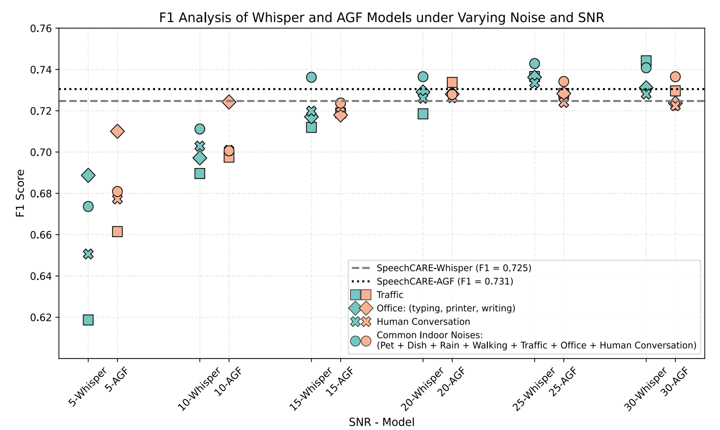


Figure 4. SpeechCARE Performance in Noisy Speech Environments



3. SpeechCARE Trustworthy: **(1) Bias Mitigation**: The SpeechCARE Framework includes the SpeechCURA toolkit to mitigate demographic bias and enhance robustness to real-world conditions such as background noise and long-term patient monitoring. Bias mitigation is achieved through a three-stage process: **(a) Pre-processing**, which expands the training data for under-represented groups using: (i) *Spectrogram augmentation*[16], introducing variability in frequency and timing to simulate differences in speech; (ii) *Synthetic speech generation* using *LLaMA-3-8B* to expand the transcript, and Spark-TTS[17] (a Text-To-Speech model) to convert the LLaMA-generated text to speech by learning the speaker's voice patterns; (iii) *Voice conversion using FreeVC*[18] (a voice conversion model) to transform the acoustic profile of a source speaker to resemble a target speaker. **(b) In-processing methods** (adversarial network[19], dynamic re-weighting[20]), which intensify penalization of misclassification for under-represented groups. **(c) Post-processing methods** (calibration[21], threshold optimization[22]), which balance sensitivity and specificity. **PERFORMANCE AND IMPLICATIONS.** On the Phase-2 dataset, using FreeVC voice conversion achieved the greatest bias reduction (**Figure 3**), and other methods only yielded marginal improvements. This result is driven by our Speaker-Pair-Selection method: each speaker is encoded as a vector of shimmer, mean MFCCs, and speech rate—features linked to cognitive status. Cosine distances are computed between all pairs, and maximum-weight matching selects disjoint pairs with maximal acoustic dissimilarity. Training on these high-contrast pairs outperformed random speaker pairing. Overall, no single bias mitigation strategy is effective across all datasets. The SpeechCURA framework enables systematic testing and selection of optimal bias mitigation methods tailored to specific datasets and use cases.

(2) Robustness–Noise Tolerance: To assess noise robustness, background sounds (e.g., traffic, conversation, rain, **Figure 4**) were added to Phase-2 dataset. SpeechCARE-AGF outperformed SpeechCARE-Whisper across noise types, due to its AGF fusion and gating network, which down-weights noisy acoustic inputs in favor of stable transcript inputs. Fine-tuning on noisy data by adding noise to the training set improved F1 scores by 5.5%, demonstrating resilience to noisy environments. **(3) Speaker verification** is essential for longitudinal studies where patients submit speech remotely. Each incoming sample is matched to a known voiceprint using the "Pyannote-audio[23]" toolkit. To evaluate accuracy, we used the Delaware corpus, where each speaker completed multiple recordings. With an optimized threshold (cosine distance $\leq$ 0.33), we achieved 100% accuracy in linking samples to the correct individual.

IMPLICATIONS. SpeechCARE's fusion network demonstrates greater robustness to noise than large transformer models like Whisper, benefiting from its ability to down-weight noisy inputs. For longitudinal home care, preserving patient identity is critical; SpeechCURA addresses this via accurate speaker verification, enhancing privacy and continuity.

4. SpeechCARE Generalizability on DementiaBank. SpeechCARE models were evaluated on five DementiaBank corpora to assess generalizability (Table 1). WLS[24] (English), Lu[25] (Mandarin), and PerLA[26] (Spanish) had no participant overlap with the Phase-2 dataset and included only one diagnostic group (**Table 1**). In corpora with control and MCI samples (Delaware and Chou, **Table 1**), test sets (20%) were split in a way that avoids any speaker overlap with the Phase-2 dataset. On these corpora, we evaluated SpeechCARE-Whisper and AGF using four strategies: **(1) Zero-shot inference (Direct application)** – We tested the existing SpeechCARE models on new data without any changes; **(2) Transfer learning**–fine-tuning the Phase-2-trained model on the target corpus; **(3) In-corpus training**–skipping Phase-2 data, we trained models using only the Chou or Delaware data, to explore the value of longer recordings (>30s) and having multiple speech samples per participant. **(4) Adaptive-Task Attention (Task-weighted modeling).** We developed a method that learns which speech tasks (e.g., story recall vs. picture description) are most informative for detecting MCI. It performed well on both Delaware and Chou (Table 1) and revealed task types valuable for screening.

Table 1. Dataset Characteristics and Generalizability Results of SpeechCARE on DementiaBank Corpora

Corpus	Participants	Language/Speech Tasks	Overlap with Phase-2 Dataset	Strategy of Evaluation	Best Performing Model	Performance
WLS	1,368 HC	English/Cookie-Theft Picture	832; 538 reserved	Zero-shot Inference	AGF	F1= 0.99
Lu	51 ADRD	Mandarin/Cookie-Theft Picture	None	Zero-shot Inference	AGF	F1= 0.98
PerLA	21 ADRD	Spanish/Personal Narrative	None	Zero-shot Inference	AGF	F1= 0.96
Chou	40 HC, 47 MCI	Mandarin/ Picture description: Daddy, Park, Night-market	None	Zero-shot Inference Transfer learning In-corpus training Task-Attentive	AGF AGF AGF AGF	F1= 0.09 F1= 0.84 F1= 0.92 F1= 0.82
Delaware	106 HC, 99 MCI*	English/ Picture description: Cookie Theft, Cat Rescue, Rockwell; Story recall: Cinderella, Procedural Narrative: PB&J Sandwich	81 participants; 113 recordings (some recorded two sessions)	Zero-shot Inference Transfer learning In-corpus training Task-Attentive	AGF AGF AGF AGF	F1=0.41 F1=0.70 F1= 0.78 F1=0.78

* A pair of duplicate IDs with inconsistent labels was removed from the data

Figure 5.A Task-level and participant-level performance of SpeechCARE- AGF on the Chou dataset.

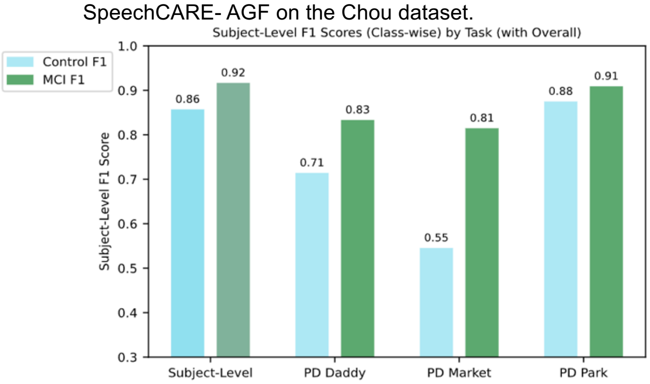
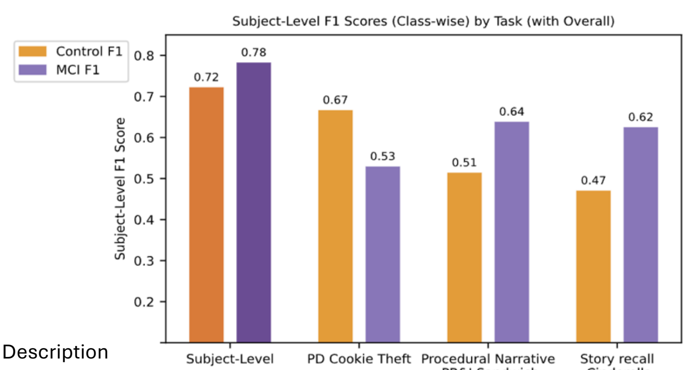


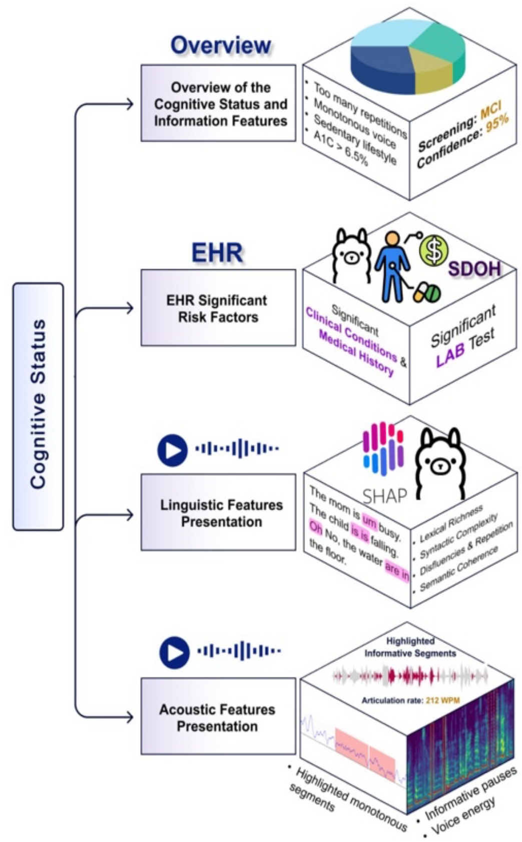
Figure 5.B Task-level and participant-level performance of SpeechCARE- AGF on the Delaware dataset.



***PD**: Picture Description

IMPLICATIONS. (1) Multilingual applicability: SpeechCARE generalized across English, Mandarin, and Spanish with minimal language-specific tuning. (2) Short recordings are effective: ≤60s samples (45s in Chou; 15–30s in Delaware) captured impairment in phonetic motor planning[27] (e.g., distorted phonemes, articulatory inconsistency). (3) Task selection matters: Delaware tasks like Cookie Theft, story recall, procedural narrative (**Figure 5B**), and Chou's tasks like PD-Park and PD-Daddy (**Figure 5A**) showed the highest MCI discrimination. While Cat-Rescue and Rockwell from Delaware did not contribute to MCI detection, we removed them from the final analysis. Multiple, diverse tasks improved sensitivity and

Figure 6. Overview of Explainability Framework



specificity by capturing broader deficits. (4) No universal model: Optimal performance requires testing multiple transformers, integrating other modalities (e.g., clinical data), and using fusion strategies to boost accuracy across datasets.

5. Explainability Framework (Figure 6). For Phase-3, we conducted usability interviews with clinicians at Columbia Medical Center (IRB # AAAV8664), which helped refine the explainability framework introduced in Phase-2. Specifically, we implemented: **(1) Overview panel: Summary of important findings**, including (a) model outcome and confidence score; (b) the four most influential clinical and speech risk factors; (c) the relative contribution of each data modality to the model's decision. **(2) Expandable interpretation modules**, including: (a) *Clinical Biomarkers*: identify important Electronic Health Records (EHR)-derived clues (e.g., clinical: GAD >10, behavioral: sleep disruption, social: social isolation) using LLaMA-3.3-70B, trained with expert prompting and Retrieval-Augmented Generation[28] over structured EHR data, enabling contextualization of speech predictions within broader health profiles; (b) *Linguistic Biomarkers*: combine SHAP[29] token-level attribution and LLaMA-3.3-70B (trained with Tree-of-Thought[30] reasoning) to highlight key linguistic cues and provide structured interpretation across four domains: lexical diversity, fluency, syntactic structure, and semantic coherence; (c) *Acoustic Biomarkers*: integrate mHuBERT gradient saliency maps[31] to highlight informative waveform segments (e.g., high instability); SHAP-annotated spectrograms to visualize frequency-energy patterns, along with plotting fundamental frequency (vocal fold vibration) and informative pauses (e.g., pre-noun pauses); Rhythmic structure is analyzed using Shannon entropy[32] to detect monotonous segments of voice. **Usability Evaluation (IRB # AAAV8664):** Stage 1: We conducted think-aloud interviews with clinicians. Recordings were analyzed to guide UI revisions. Clinicians were unfamiliar with spectrograms, so we are developing clear guidelines. Stage 2: We are refining the interface to meet usability targets (system usability score ≥ 80, Health-ITUES ≥ 4.5). The dashboard prototype, codebooks, examples, and evaluation protocol have been submitted with this report. **IMPLICATIONS.** Clinicians expressed strong interest in the explainability framework, noting its potential to enhance understanding of speech biomarkers and clinical risk factors associated with ADRD.

6. SpeechCARE Multi-Task Learning (Figure 7). Many acoustic and linguistic markers of ADRD overlap with other neurological or physiological conditions. Leveraging this overlap, Multi-Task Learning[33] (MTL) offers a promising solution by enabling models to learn shared patterns across related conditions, reducing false-positive rates and improving generalizability. To address this, we implemented a multi-task learning (MTL) pipeline using 8,029 speech samples from five datasets: Cognitive Impairment (phase-2 dataset), Aphasia, Emotion[34], Parkinson[35], and COVID-19[36] (**Figure 7)** for sample size). **Model Architecture and Results.** The SpeechCARE-MTL was built on two architectures (AGF and Whisper) and each included a shared feature network, task-specific output heads, an adversarial domain discriminator (that promotes task-invariant representations), and Homoscedastic Uncertainty[37] for loss balancing. SpeechCARE-Whisper consistently showed improved performance when trained with auxiliary tasks. ADRD F1 scores were highest with AphasiaBank (0.7379) and lowest with UK Covid-19 dataset (0.7233). The corresponding F1 scores for the auxiliary tasks were

Figure 7. Multi-Task Learning with

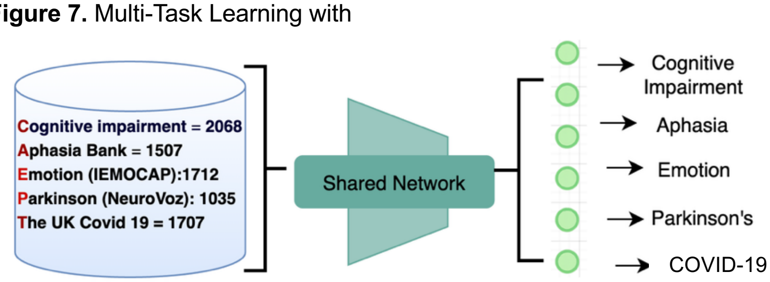
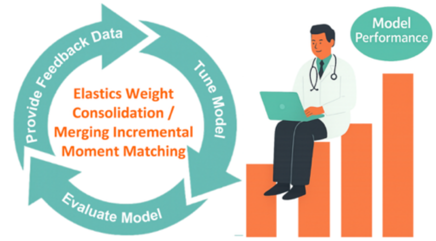
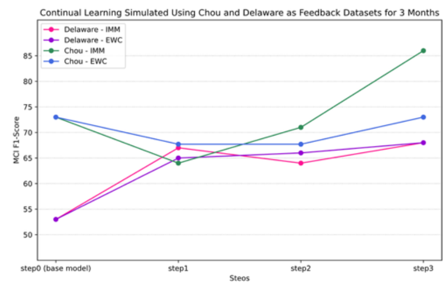


Figure 8. Continual learning



0.8349 (aphasia detection), 0.6187 (emotion detection), 0.7561 detection), and 0.6044 (COVID-19 detection). **IMPLICATIONS.** This to apply MTL across clinically overlapping speech datasets to improve specificity. By leveraging auxiliary tasks, MTL enables the model to discriminative and generalizable patterns. Shared representation tasks reduces overfitting, enhances robustness, and mitigates diagnostic making the model more suitable for real-world screening.

Figure 9. Continual learning Results



7. Continual Learning (Figure 8). ADRD speech models, trained on curated corpora, often face new patient demographics and clinician feedback after deployment. In such situations, continual learning[39] (CL) helps models adapt while preserving prior knowledge. To evaluate this, we applied two CL algorithms: Elastic Weight Consolidation[40] (EWC) and Incremental Moment Matching[41] (IMM). EWC uses the diagonal Fisher Information Matrix[42] with a quadratic penalty to constrain changes to task-critical weights, allowing integration of clinician labels without forgetting. IMM treats successive parameter sets as Gaussian posteriors, fusing them via mean-IMM or Fisher-weighted mode-IMM to provide lightweight Bayesian updates that mitigate catastrophic forgetting. **Experiments.** We simulated two deployments for SpeechCARE: an English clinic with the Delaware dataset (n =205, Table 1) and a Mandarin clinic with the Chou dataset (n =87, Table 1). Each setting assumed a three-month deployment period, where clinician feedback was simulated by randomly selecting one subset of training data per month. Across three rounds, models were incrementally updated using either EWC or IMM, treating each subset as hypothetical feedback. Performance was evaluated after each update using the same holdout test sets from the earlier generalizability study in section 4 (SpeechCARE Generalizability on DementiaBank). **RESULTS and IMPLICATIONS.** As shown in **Figure 9**, IMM consistently outperformed EWC across both settings, and both improved SpeechCARE's performance over three months. This is the first study to demonstrate the utility of these methods in ADRD research, showing that models can learn from ongoing clinician feedback to enhance MCI detection.

Figure 10.a structured speech in patient home

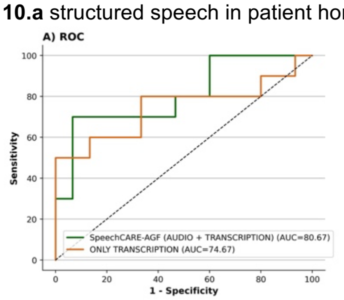


Figure 10.b Semi-structured interview over phone

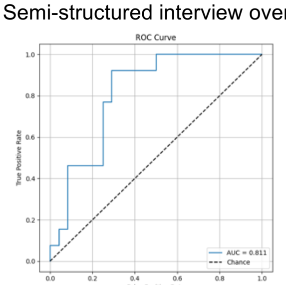
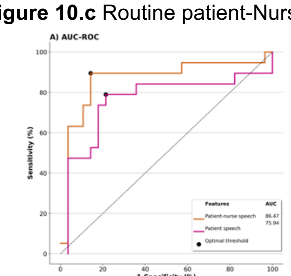


Figure 10.c Routine patient-Nurse conversation



8. SpeechCARE. Application in Real-world Care Settings. Supported by the NIH R00 (AG076808) and A2 Collective (P30-AG-073105), we expanded the application of SpeechCARE in home healthcare to enable early detection of MCI across diverse care contexts. Of 443 patients contacted, 135 agreed to participate; 113 completed in-home assessments, 98 participated in phone calls, and 47 were involved in patient–nurse recorded encounters. (1) **In-home RA–patient conversations** (**Figure 10.a**), collected during structured cognitive assessments, captured linguistic impairments, yielding an AUC=80.76 and an F1=70.6%. (2) **RA–patient phone calls** (**Figure 10.b**), conducted 3–5 weeks post-assessment, provided spontaneous speech reflecting recall ability and verbal organization, which achieved an AUC=81.13 and an F1=72.7%. (3) **Patient–nurse conversations during routine visits (Figure 10.c)**, revealed rich linguistic and interactional markers of cognitive decline (e.g., repetition, turn-taking gaps), achieving the highest accuracy with an AUC=86.47 and an F1= 85%**. IMPLICATIONS.** This is the first study to evaluate cognitive-speech signals across three naturalistic home care speech contexts. The approach enables passive and scalable MCI screening through integration into real-world workflows.

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